

Instructions for AGT 2025

- Each part should be completed in groups of 3 to 4 members. If you wish to work individually on all parts, please contact me.
- All sections, except for the “Scribe” section, should be written in English. Submissions in Chinese will receive a grade of zero.

Homeworks (20*2)

File name: “group_name+hw1/2.pdf”

Presentation (5)

Prepare several PowerPoint slides covering the background of the research paper, the reasons your group chose it, and your group’s project schedule.

File name: “group_name+pre.pptx”

Final Report (35)

- Grading for the projects will be based on *sincerity* with which the project was written and how much useful *information* I can get from.
- You may include any relevant information about the research paper, such as figures, tables, or explanations, to help the reader understand the content.
- The following sections should be included: introduction, technical overview/intuition/your thoughts, a few results you find interesting (or an attempt to simplify the proof in your own words, optional), the difficult parts you couldn't understand, and the approaches you have tried, finally, summary (even for the whole class..)

File name: “group_name+final.pdf”

Scribe (20)

I recommend using Chinese for the scribe (English if you are an international student). You may use LaTeX, Word, or Markdown to edit the content. However, please submit the final version as a PDF, along with all the necessary source files required for compilation.

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1. Feldman, Michal, Jugal Garg, Vishnu V. Narayan, and Tomasz Ponitka. "Proportionally Fair Makespan Approximation." *arXiv preprint arXiv:2412.08572* (2024).
 2. Garg, Jugal, Xin Huang, and Erel Segal-Halevi. "Improved Maximin Share Approximations for Chores by Bin Packing." *arXiv preprint arXiv:2411.04391* (2024).
 3. Garg, Jugal, Aniket Murhekar, and John Qin. "Weighted EF1 and PO allocations with few types of agents or chores." *arXiv preprint arXiv:2402.17173* (2024).
 4. Feng, Yuda, Yang Hu, Shi Li, and Ruilong Zhang. "Constant Approximation for Weighted Nash Social Welfare with Submodular Valuations." *arXiv preprint arXiv:2411.02942* (2024).
 5. Charikar, Moses, Alexandra Lassota, Prasanna Ramakrishnan, Adrian Vetta, and Kangning Wang. "Six Candidates Suffice to Win a Voter Majority." *arXiv preprint arXiv:2411.03390* (2024).
 6. Feng, Yiding, and Yaonan Jin. "Beyond Regularity: Simple versus Optimal Mechanisms, Revisited." *arXiv preprint arXiv:2411.03583* (2024).
 7. Akrami, Hannaneh, Jugal Garg, Eklavya Sharma, and Setareh Taki. "Improving approximation guarantees for maximin share." *arXiv preprint arXiv:2307.12916* (2023).
 8. Charikar, Moses, Prasanna Ramakrishnan, Kangning Wang, and Hongxun Wu. "Breaking the metric voting distortion barrier." *Journal of the ACM* 71, no. 6 (2024): 1-33.
 9. Chen, Xi, and Yuhao Li. "Constant Inapproximability of Pacing Equilibria in Second-Price Auctions." *arXiv preprint arXiv:2501.15295* (2025).

10. Brânzei, Simina, Reed Phillips, and Nicholas Recker. "Tarski Lower Bounds from Multi-Dimensional Herringbones." arXiv preprint arXiv:2502.16679 (2025).
11. Chen, Thomas, Xi Chen, Binghui Peng, and Mihalis Yannakakis. "Computational Hardness of the Hylland-Zeckhauser Scheme*." In Proceedings of the 2022 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA), pp. 2253-2268. Society for Industrial and Applied Mathematics, 2022.
12. Deligkas, Argyrios, Eduard Eiben, Robert Ganian, Tiger-Lily Goldsmith, and Stavros D. Ioannidis. "The Complexity of Extending Fair Allocations of Indivisible Goods." arXiv preprint arXiv:2503.01368 (2025).
13. Deligkas, Argyrios, John Fearnley, Alexandros Hollender, and Themistoklis Melissourgios. "Constant inapproximability for Fisher markets." In Proceedings of the 25th ACM Conference on Economics and Computation, pp. 13-39. 2024.
14. Yao, Andrew Chi-Chih. "Dominant-strategy versus bayesian multi-item auctions: Maximum revenue determination and comparison." In Proceedings of the 2017 ACM Conference on Economics and Computation, pp. 3-20. 2017.
15. Duan, Ran, and Kurt Mehlhorn. "A combinatorial polynomial algorithm for the linear Arrow-Debreu market." In International Colloquium on Automata, Languages, and Programming, pp. 425-436. Berlin, Heidelberg: Springer Berlin Heidelberg, 2013.
16. Khot, Subhash, Richard J. Lipton, Evangelos Markakis, and Aranyak Mehta. "Inapproximability results for combinatorial auctions with submodular utility functions." Algorithmica 52 (2008): 3-18.
17. Chekuri, Chandra, Pooja Kulkarni, Rucha Kulkarni, and Ruta Mehta. "1/2-Approximate MMS Allocation for Separable Piecewise Linear Concave Valuations." In Proceedings of the AAAI Conference on Artificial Intelligence, vol. 38, no. 9, pp. 9590-9597. 2024.
18. Daskalakis, Constantinos, and Seth Matthew Weinberg. "Symmetries and optimal multi-dimensional mechanism design." In Proceedings of the 13th ACM conference on Electronic commerce, pp. 370-387. 2012.
19. Garg, Jugal, Ruta Mehta, Vijay V. Vazirani, and Sadra Yazdanbod. "Settling the complexity of Leontief and PLC exchange markets under exact and approximate equilibria." In Proceedings of the 49th Annual ACM SIGACT Symposium on Theory of Computing, pp. 890-901. 2017.
20. Bei, Xiaohui, Yuda Feng, Yang Hu, Shi Li, and Ruilong Zhang. "Nash Social Welfare with Submodular Valuations: Approximation Algorithms and Integrality Gaps." arXiv preprint arXiv:2504.09669 (2025).
21. Guruganesh, Guru, Aranyak Mehta, Di Wang, and Kangning Wang. "Prior-Independent Auctions for Heterogeneous Bidders." In Proceedings of the 2024 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA), pp. 1-18. Society for Industrial and Applied Mathematics, 2024.
22. Führlich, Pascal, Ágnes Cseh, and Pascal Lenzner. "Improving ranking quality and fairness in Swiss-system chess tournaments." In Proceedings of the 23rd ACM Conference on Economics and Computation, pp. 1101-1102. 2022.
23. Filippas, Apostolos, John J. Horton, M. I. T. Fordham, NBER Elliot Lipnowski Prasanna Parasurama, and Yale Emory. "The Production and Consumption of Social Media." (2025).
24. Wang, Hao, and Xiaohui Bei. "Real-time driver-request assignment in ridesourcing." In Proceedings of the AAAI conference on artificial intelligence, vol. 36, no. 4, pp. 3840-3849. 2022.
25. Pycia, Marek, and Kyle Woodward. "Pollution Permits: Efficiency by Design." Available at SSRN 4453956 (2023).
26. Procaccia, Ariel, Isaac Robinson, and Jamie Tucker-Foltz. "School redistricting: Wiping unfairness off the map." In Proceedings of the 2024 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA), pp. 2704-2724. Society for Industrial and Applied Mathematics, 2024.
27. Tomlinson, Kiran, Tanvi Namjoshi, Johan Ugander, and Jon Kleinberg. "Replicating electoral success." In Proceedings of the AAAI Conference on Artificial Intelligence, vol. 39, no. 13, pp. 14167-14175. 2025.
28. Immorlica, Nicole, Nicholas Wu, and Brendan Lucier. "Maximal Procurement under a Budget." arXiv preprint arXiv:2404.15531 (2024).
29. Harris, Keegan, Nicole Immorlica, Brendan Lucier, and Aleksandrs Slivkins. "Algorithmic persuasion through simulation." arXiv preprint arXiv:2311.18138 (2023).

30. Paccagnan, Dario, and Martin Gairing. "In congestion games, taxes achieve optimal approximation." In Proceedings of the 22nd ACM Conference on Economics and Computation, pp. 743-744. 2021.
31. Chen, Houshuang, Yaonan Jin, Pinyan Lu, and Chihao Zhang. "Tight Regret Bounds for Fixed-Price Bilateral Trade." arXiv preprint arXiv:2504.04349 (2025).
32. Dütting, Paul, Federico Fusco, Silvio Lattanzi, Ashkan Norouzi-Fard, Ola Svensson, and Morteza Zadimoghaddam. "The Cost of Consistency: Submodular Maximization with Constant Recourse." arXiv preprint arXiv:2412.02492 (2024).
33. Cesa-Bianchi, Nicolò, Tommaso Cesari, Roberto Colomboni, Federico Fusco, and Stefano Leonardi. "The role of transparency in repeated first-price auctions with unknown valuations." In Proceedings of the 56th Annual ACM Symposium on Theory of Computing, pp. 225-236. 2024.
34. Dütting, Paul, Tomer Ezra, Michal Feldman, and Thomas Kesselheim. "Multi-agent combinatorial contracts." In Proceedings of the 2025 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA), pp. 1857-1891. Society for Industrial and Applied Mathematics, 2025.
35. Garg, Jugal, Aniket Murhekar, and John Qin. "Fair division of indivisible chores via earning restricted equilibria." arXiv preprint arXiv:2407.03318 (2024).
36. Garg, Jugal, Yixin Tao, and László A. Végh. "Approximating Competitive Equilibrium by Nash Welfare." In Proceedings of the 2025 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA), pp. 2538-2559. Society for Industrial and Applied Mathematics, 2025.